

§7.1 - Integration by Parts

Definition #1 (Real numbers)

The set of all **real numbers** is denoted by

$$\mathbf{R}.$$

If x is a real number we express this as

$$x \in \mathbf{R}.$$

Definition #2

We also have the sets of **natural numbers**, **integers**, and **rational numbers**:

$$\mathbf{N} = \{1, 2, 3, \dots\}$$

$$\mathbf{Z} = \{0, \pm 1, \pm 2, \pm 3, \dots\}$$

$$\mathbf{Q} = \{m/n \mid m \in \mathbf{Z} \text{ and } n \in \mathbf{N}\}.$$

Observe

$$\mathbf{N} \subset \mathbf{Z} \subset \mathbf{Q} \subset \mathbf{R}.$$

We have

$$-3 \in \mathbb{Z}, \quad -\frac{3}{2} \in \mathbb{Q}, \quad \text{and} \quad \sqrt{2} \in \mathbb{R}$$

but

$$-3, -\frac{3}{2}, \sqrt{2} \notin \mathbb{N}.$$

Definition #3 (Antiderivative)

Let F and f be functions on an interval I .

If

$$F'(x) = f(x)$$

on I then F is an **antiderivative** of f . We denote this as

$$\int f(x) \, dx = F(x)$$

Theorem #1

Let F be any antiderivative of the function f on an interval I .

Then all of f 's antiderivatives are of the form

$$F(x) + C$$

where $C \in \mathbf{R}$.

We know

$$\int e^x dx = e^x + C$$

while

$$\int x^5 dx = \frac{1}{6}x^6 + C$$

where $C \in \mathbb{R}$.

Theorem #2 (Fundamental Theorem of Calculus)

Let f be continuous on the interval $[a, b]$ and define g by

$$g(x) = \int_a^x f(t) \, dt.$$

Then

1. g is continuous on $[a, b]$ and g' exists on (a, b) where

$$g'(x) = f(x)$$

2. g is an antiderivative of f on $[a, b]$.

3. We have

$$F(b) - F(a) = \int_a^b f(x) \, dx$$

where F is any antiderivative of f on $[a, b]$.

Remark

Some shorthand:

$$F(b) - F(a) = F(x)]_a^b = [F(x)]_a^b = F(x)|_{x=a}^b$$

Theorem #3 (Chain Rule)

Let g be differentiable at x and f be differentiable at $g(x)$.

Then

$$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x).$$

Theorem #4 (Integration by Substitution)

Let $u = g(x)$ be a differentiable function whose range is an interval I .

If f is continuous on I then

$$\int f(g(x)) \cdot g'(x) \, dx = \int f(u) \, du.$$

We have

$$\frac{d}{dx} e^{-x^2} = e^{-x^2} \frac{d}{dx} (-x^2) = e^{-x^2} (-2x)$$

so

$$\int -2xe^{-x^2} dx = e^{-x^2} + C.$$

Instead, if we apply a substitution

$$u = -x^2 \quad du = -2x dx$$

we have

$$\int -2xe^{-x^2} dx = \int e^u du = e^u + C = e^{-x^2} + C.$$

Example #1

Find

$$\int \frac{x}{x-3} dx.$$

Let

$$u = x - 3 \quad du = dx$$

so $x = u + 3$. We then have

$$\begin{aligned} \int \frac{x}{x-3} dx &= \int \frac{u+3}{u} du \\ &= \int 1 + \frac{3}{u} du \\ &= u + 3 \ln|u| + C. \end{aligned}$$

If we back substitute for x we find

$$\begin{aligned} \int \frac{x}{x-3} dx &= x - 3 + 3 \ln|x - 3| + C \\ &= x + 3 \ln|x - 3| + C \end{aligned}$$

where $C \in \mathbb{R}$. Notice that " $C - 3$ " = C .

Theorem #5 (Integration by Substitution for Definite Integrals)

Let g' be continuous on the interval $[a, b]$.

If f is continuous on the range of $u = g(x)$ then

$$\int_a^b f(g(x)) \cdot g'(x) \, dx = \int_{g(a)}^{g(b)} f(u) \, du.$$

Theorem #6 (Product Rule)

Let u and v be differentiable functions.

Then

$$\frac{d}{dx} (u(x) \cdot v(x)) = u'(x) \cdot v(x) + u(x) \cdot v'(x).$$

By the Product Rule

$$\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$$

and so

$$f(x)g(x) = \int f'(x)g(x) dx + \int f(x)g'(x) dx.$$

We have

$$\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx$$

and let

$$u = f(x), \quad v = g(x), \quad du = f'(x) dx, \quad dv = g'(x) dx$$

then

$$\int u dv = u \cdot v - \int v du.$$

Theorem #7 (Integration by Parts)

Let u and v be differentiable functions.

Then

$$\int u \, dv = u \cdot v - \int v \, du$$

Example #2

Find

$$\int 3xe^x \, dx$$

Let

$$u = 3x \quad dv = e^x dx$$

$$du = 3dx \quad v = e^x$$

then by Integration by Parts

$$\int 3xe^x dx = \int u dv$$

$$= uv - \int v du$$

$$= 3xe^x - \int 3e^x dx$$

$$= 3xe^x - 3e^x + C$$

$$= 3(x-1)e^x + C.$$

Theorem #8 (Integration by Parts for Definite Integrals)

For u' and v' continuous,

$$\int_a^b u(x) \cdot v'(x) \, dx = u(x) \cdot v(x) \Big|_{x=a}^b - \int_a^b v(x) \cdot u'(x) \, dx.$$

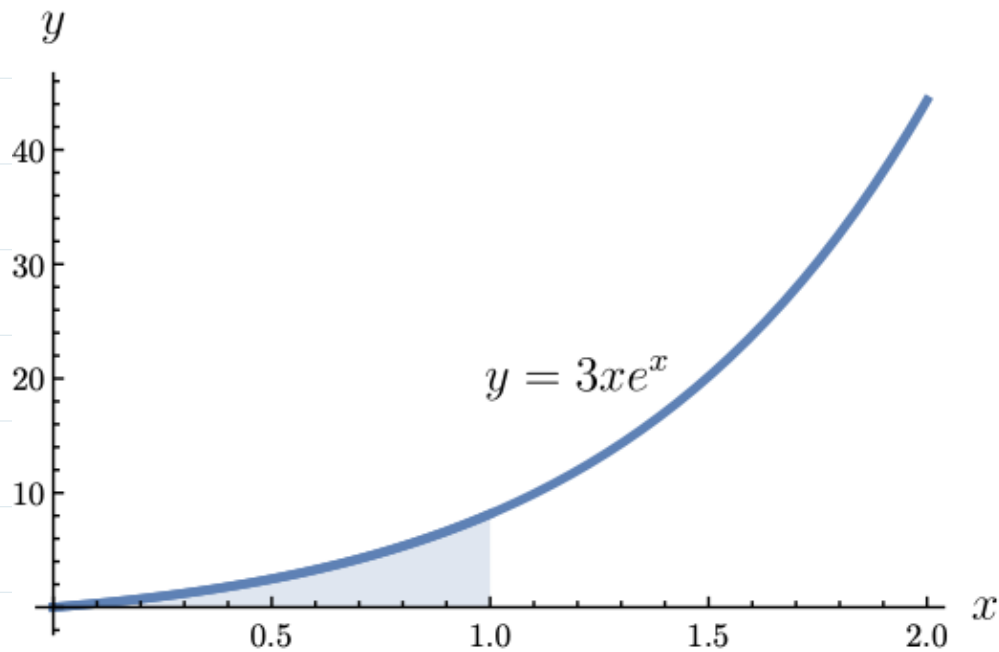
Example #3

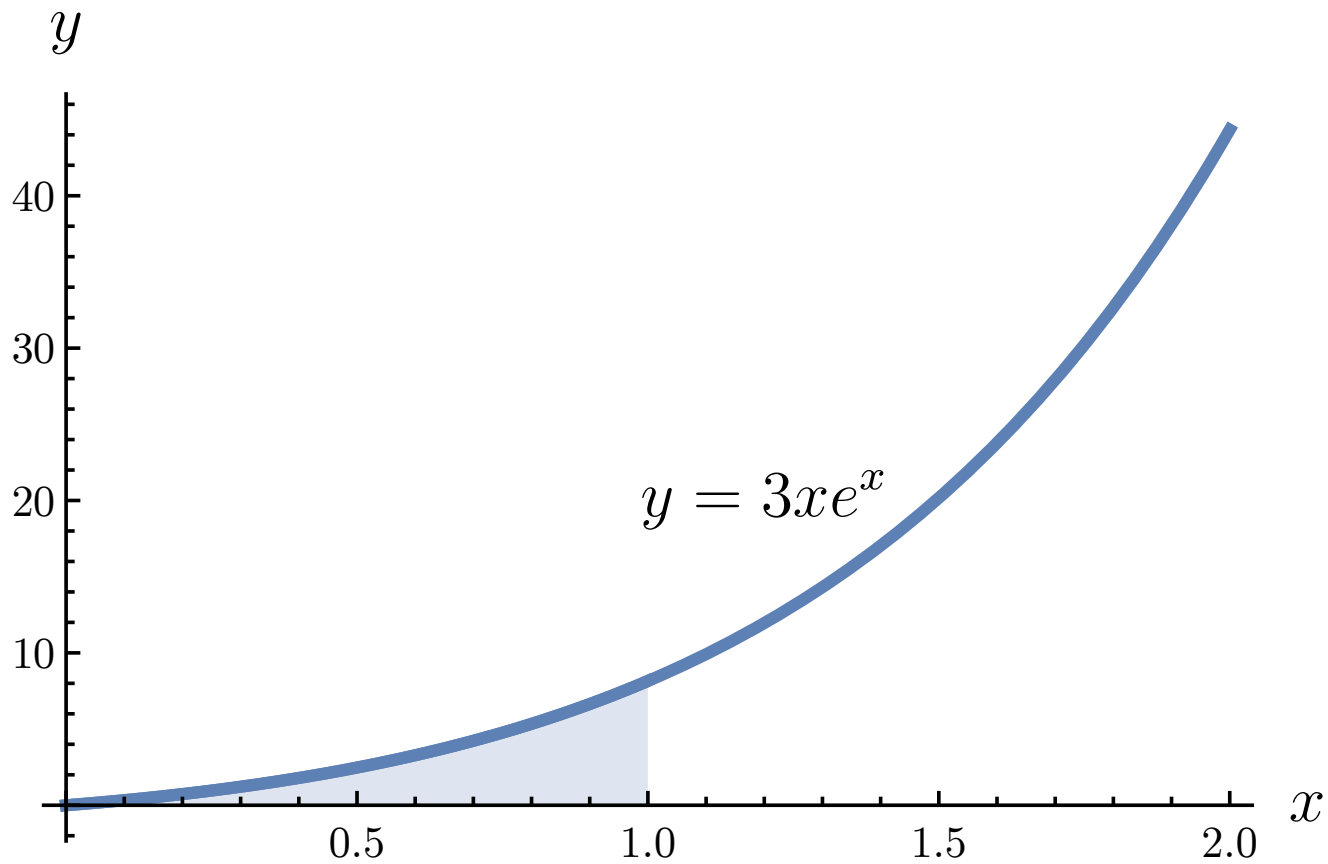
Find

$$\int_0^1 3xe^x \, dx.$$

By the Fundamental Theorem of Calculus,

$$\begin{aligned}\int_0^1 3xe^x dx &= 3(x-1)e^x \Big|_{x=0}^1 \\ &= 3 \cdot (1-1) \cdot e - 3 \cdot (0-1) \cdot 1 \\ &= 3.\end{aligned}$$





$$\begin{array}{ccc}
 u's & & dv's \\
 \hline
 & & \\
 u & \xrightarrow{+} & dv \\
 du & \xrightarrow{-} & v
 \end{array}$$

maps to

$$\int u \, dv = u \cdot v - \int v \, du$$

Example #4

Find

$$\int \ln(x) \, dx.$$

There is no product of functions!

Let

$$u = \ln(x) \quad dv = 1 \cdot dx$$

$$du = \frac{1}{x} dx \quad v = x$$

then

$$\int \ln(x) \cdot 1 \, dx = \ln(x) \cdot x - \int x \cdot \frac{1}{x} \, dx$$

$$= \ln(x) \cdot x - \int 1 \, dx$$

$$= \ln(x) \cdot x - x + C$$

Example #5

Find

$$\int x^5 \ln(x) \, dx.$$

Let

$$u = \ln(x) \quad dv = x^5 \cdot dx$$

$$du = \frac{1}{x} dx \quad v = \frac{x^6}{6}$$

then by Integration by Parts

$$\int x^5 \ln(x) dx = \int u dv$$

$$= uv - \int v du$$

$$= \ln(x) \cdot \frac{x^6}{6} - \int \frac{x^6}{6} \cdot \frac{1}{x} dx$$

$$= \frac{x^6}{6} \ln(x) - \int \frac{x^5}{6} dx$$

$$= \frac{x^6}{6} \ln(x) - \frac{x^6}{36} + C$$

where $C \in \mathbb{R}$.

Remark

If we have

$$\int p(x) \ln(x) \, dx$$

where p is a polynomial then choose

$$u = \ln(x) \quad \text{and} \quad dv = p(x) \, dx.$$

So differentiate the logarithm.

Example #6

Find

$$\int x^2 e^{2x} \, dx.$$

Let

$$u = x^2 \quad dv = e^{2x} dx$$
$$du = 2x dx \quad v = \frac{1}{2}e^{2x}$$

then by Integration by Parts

$$\int x^2 e^{2x} dx = \int u dv$$

$$= uv - \int v du$$

$$= x^2 \cdot \frac{1}{2}e^{2x} - \int 2x \cdot \frac{1}{2}e^{2x} dx$$

$$= \frac{1}{2}x^2 e^{2x} - \underbrace{\int x e^{2x} dx}$$

more integration
by parts!

Rather than apply the process again, and introduce confusion, we can use a table:

$u's$		$dv's$
x^2	$+$	e^{2x}
$2x$	$-$	$\frac{1}{2}e^{2x}$
2	$+$	$\frac{1}{4}e^{2x}$
0		$\frac{1}{8}e^{2x}$

Thus

$$\begin{aligned}\int x^2 e^{2x} dx &= \left(x^2 \cdot \frac{1}{2} - 2x \cdot \frac{1}{4} + 2 \cdot \frac{1}{8} \right) e^{2x} + C \\ &= \left(\frac{x^2}{2} - \frac{x}{2} + \frac{1}{4} \right) e^{2x} + C\end{aligned}$$

where $C \in \mathbb{R}$.

Example #7

Find

$$\int x^5 e^x \, dx.$$

$u's$		$dv's$
x^5	+	e^x
$5x^4$	-	e^x
$20x^3$	+	e^x
$60x^2$	-	e^x
$120x$	+	e^x
120	-	e^x
0		e^x

Thus

$$\int x^5 e^x dx$$

$$= (x^5 - 5x^4 + 20x^3 - 60x^2 + 120x - 120)e^x + C$$

where $C \in \mathbb{R}$.

Remark

If we have

$$\int p(x)e^{kx} dx$$

where p is a polynomial then choose

$$u = p(x) \quad \text{and} \quad dv = e^{kx} dx.$$

So differentiate the polynomial.

Example #8

Find

$$\int x^2 \sin(x) \, dx$$

Let

$$u = x^2 \quad dv = \sin(x) \, dx$$
$$du = 2x \, dx \quad v = -\cos(x)$$

then by Integration by Parts

$$\int x^2 \sin(x) \, dx = \int u \, dv$$

$$= uv - \int v \, du$$

$$= x^2(-\cos(x)) - \int 2x \cdot (-\cos(x)) \, dx$$

$$= -x^2 \cos(x) + \int 2x \cos(x) \, dx$$

more integration
by parts!

Rather than apply the process again, and introduce confusion, we can use a table:

u's		dv's
x^2	+	$\sin(x)$
$2x$	-	$-\cos(x)$
2	+	$-\sin(x)$
0		$\cos(x)$

Thus

$$\int x^2 \sin(x) dx$$

$$= x^2(-\cos(x)) - 2x(-\sin(x)) + 2\cos(x) + C$$

$$= (-x^2 + 2)\cos(x) + 2x\sin(x) + C$$

where $C \in \mathbb{R}$.

Remark

If we have

$$\int p(x) \sin(kx) \, dx \quad \text{or} \quad \int p(x) \cos(kx) \, dx$$

where p is a polynomial then choose

$$u = p(x) \quad \text{and} \quad dv = \sin(kx) \, dx \quad \text{or} \quad dv = \cos(kx) \, dx.$$

So differentiate the polynomial.

Example #9

Find

$$\int \sin(x) e^{2x} \, dx.$$

Apply integration by parts twice and generate a multiple of

$$\begin{array}{rcl}
 & \int u \, dv & \\
 \hline
 u's & & dv's \\
 \hline
 \sin(x) & \xrightarrow{+} & e^{2x} \\
 \cos(x) & \xrightarrow{-} & \frac{1}{2}e^{2x} \\
 -\sin(x) & \xrightarrow{+} & \frac{1}{4}e^{2x}
 \end{array}$$

Bottom row represents $\int v \, du$

$$\frac{\int \sin(x)e^{2x} \, dx}{I}$$

$$= \left(\frac{1}{2}\sin(x) - \frac{1}{4}\cos(x) \right) e^{2x} - \frac{1}{4} \frac{\int \sin(x)e^{2x} \, dx}{I}$$

Solve for I :

$$I = \left(\frac{1}{2} \sin(x) - \frac{1}{4} \cos(x) \right) e^{2x} - \frac{1}{4} I$$

$$I + \frac{1}{4} I = \left(\frac{1}{2} \sin(x) - \frac{1}{4} \cos(x) \right) e^{2x} + C$$

$$\frac{5}{4} I = \left(\frac{1}{2} \sin(x) - \frac{1}{4} \cos(x) \right) e^{2x} + C$$

$$I = \frac{4}{5} \left(\frac{1}{2} \sin(x) - \frac{1}{4} \cos(x) \right) e^{2x} + C_1$$

Hence

$$\int \sin(x) e^{2x} dx = \left(\frac{2}{5} \sin(x) - \frac{1}{5} \cos(x) \right) e^{2x} + C_1$$

Remark

If we have

$$\int e^{kx} \sin(mx) \, dx \quad \text{or} \quad \int e^{kx} \cos(nx) \, dx$$

apply tabular integration by parts twice, and then carefully solve for the symbol

$$\int e^{kx} \sin(mx) \, dx \quad \text{or} \quad \int e^{kx} \cos(nx) \, dx.$$

It does not matter which term you choose as u or dv , as long as you are consistently choosing the same “type” of term.

Example #10

Find

$$\int \ln^2(x) \, dx = \int \ln(x) \cdot \ln(x) \, dx.$$

Let

$$u = \ln^2(x) \quad v = x$$
$$du = 2 \frac{\ln(x)}{x} dx \quad dv = dx$$

$$\int \ln^2(x) dx = x \ln^2(x) - 2 \underbrace{\int \ln(x) dx}_{\text{found earlier}}$$
$$= x \ln^2(x) - 2(\ln(x) \cdot x - x) + C$$
$$= x \ln^2(x) - 2 \ln(x) \cdot x + 2x + C$$

In general

$$\int \ln^n(x) dx = x \ln^n(x) - n \int \ln^{n-1}(x) dx$$